MAST20026 Real Analysis Semester 2 2025

Assignment 5

Due: 16 October, 11:59pm

M= a mark associated with applying an appropriate method

J= a mark associated with using correct justification

A= a mark associated with a correct answer or intermediate step

N= a mark associated with good notation or writing

Please see the assignment instructions on the Assignment 5 page on the LMS, where you accessed this file. To ease submission, please prepare assignment solutions for each question on its own page. This document is 2 pages in length.

PART A (*2 marks* total)

1 mark for demonstrating thoughtful engagement with the task.
1 mark for clearly communicated response (the response is well written, and conveys thoughts concisely).

Go to the blog article *Mathematics as an ethical practice* by Lucy Rycroft-Smith linked from the Assignment 5 page on the LMS. Read it and write a short reflective response (around half a page) that considers the following questions.

- Was there anything about the article that you found surprising?
- In what ways (if any) do you think ethics has played a part in your own journey through mathematics?
- What (if any) role do you think ethics plays in real analysis, and mathematics more generally?

PART B (*20 marks* total)

1. **More on continuity** *(8 marks)*

   (a) Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Show that if $f(x) \in \mathbb{Q}$ for all $x \in \mathbb{R}$ then f is constant.

   (b) Give an example of a subset $D \subseteq \mathbb{R}$ and a non-constant, continuous function $g : D \to \mathbb{R}$ such that
   $$g(D) = \left\{ \frac{1}{n} \mid n \in \mathbb{N}^+ \right\}$$
   or explain why no such example exists. If providing an example, briefly explain why your example is an example, highlighting any theorems required.

1M: A valid approach to solving the problem was used, such as a proof by contradiction

1M: Using the density property of rationals (can be used without stating the name "density").

1J: Stated use of intermediate value theorem.

1J: The proof is correct and essentially complete.

1N: Proof is clearly written and easy to follow

Solution.

(a) Assume for contradiction that f is not constant.
Then there exists, $a, b \in \mathbb{R}$ such that $f(a) \neq f(b)$.
Assume without loss of generality that $a < b$.
Since f is continuous on $\mathbb{R}$, f is continuous on $[a, b]$.
By density of the rationals in $\mathbb{R}$ (or results from earlier in the subject), there exists $q \notin \mathbb{Q}$ such that q is in between $f(a)$ and $f(b)$.
By the intermediate value theorem, there exists $c \in (a, b)$ such that $f(c) = q$.
Then since q is in the image of f, $q \in \mathbb{Q}$.
But we also have $q \notin \mathbb{Q}$, which is a contradiction.
Therefore f must be constant.

(b) Since we need the image to be $g(D) = \left\{\frac{1}{n} \mid n \in \mathbb{N}^+\right\}$, a good candidate function is to have:

$$D = \mathbb{N}^+, \text{ and } g(x) = \frac{1}{x}$$

The function g is clearly not constant, and the image is exactly

$$\left\{\frac{1}{n} \mid n \in \mathbb{N}^+\right\}$$

Lastly g satisfies the definition of continuity since the domain $\mathbb{N}^+$ has no limit points.

1M: A valid example is provided

1J: Clearly (and correctly) stated that g is not constant, and has the correct image

1J: Brief and correct reason provided for why g is continuous

2. **More examples** *(6 marks)*

(a) Provide an example of a sequence of functions $f_n : [-1, 1] \to \mathbb{R}$ such that (f_n) converges uniformly on $[-1, a]$ for all $a \in (-1, 1)$, but does not converge uniformly on $[-1, 1]$.

(b) Provide an example of a differentiable function on $(0, 1)$ that does not obtain its minimum on $(0, 1)$, but does obtain its maximum on $(0, 1)$.

Briefly explain why your examples are examples, highlighting any theorems required.

Solution.

(a) A number of examples are possible.
Here's one possible sequence of functions.
Let $f_n(x) : [-1, 1] \to \mathbb{R}$ be defined by

$$f_n(x) = \left(\frac{x+1}{2}\right)^n$$

1M: A valid approach to finding an example is demonstrated

1J: Demonstrated understanding of uniform convergence

1J: A brief and correct reason for why the example is not uniformly convergent on [-1,1]

1J: A biref and correct reason for why the example is uniformly convergent on [-1,a]

Then for $x \in [-1, 1)$,

$$\lim_{n\to\infty} f_n(x) = 0$$

since $\left|\frac{x+1}{2}\right| < 1$.
When $x = 1$

$$f_n(1) = 1^n = 1 \to 1, \text{ as } n \to \infty$$

So $f_n(x)$ converges pointwise to

$$f(x) = \begin{cases} 0 & \text{for } x \in [-1, 1) \\ 1 & \text{for } x = 1 \end{cases}$$

Since f is not continuous $(f_n(x))$ does not converge uniformly to $f(x)$ on $[-1, 1]$.

However let $a \in (-1, 1)$ and consider $x \in [-1, a]$.
Let $\epsilon > 0$.
Note that $x \in [-1, a] \implies 0 < |x + 1| < |1 + |a||$

$$\begin{aligned} |f_n(x) - f(x)| &= \left|\left(\frac{x+1}{2}\right)^n - 0\right| \\ &\leq \left(\frac{1+|a|}{2}\right)^n \qquad\qquad 0 < |x+1| < |1+|a|| \end{aligned}$$

Since $a \in (-1, 1)$, $\left(\frac{1+|a|}{2}\right)^n \to 0$ as $n \to \infty$. Therefore, there exists $M \in \mathbb{N}^+$ such that

$$n > M \implies \left(\frac{1+|a|}{2}\right)^n < \epsilon$$

Choosing the same M, the above calculation guarantees

$$n > M \implies |f_n(x) - f(x)| < \epsilon$$

for all $x \in [-1, a]$. So $f_n(x) \to f(x) = 0$ uniformaly on $[-1, a]$.

(b) A number of examples are possible.
Here's one possible function.
Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = -(x - \frac{1}{2})^2$.
f is a polynomial, so is differentiable.
Its maximum is at $\frac{1}{2} \in (0, 1)$.
But it does not obtain its minimum on $(0, 1)$, since f is not bounded below.

1M: A valid example is provided

1J: A brief and correct reason that the functions is differentiable, obtains its maximum on (0,1) but does not obtain its minimum on (0,1)

Remark: Note that the question does not require the domain to be $(0, 1)$, so the above example is valid. Examples where the domain is $(0, 1)$ can work as well.

3. **Don't forget the +C** *(6 marks)* Let $F : \mathbb{R} \to \mathbb{R}$, $G : \mathbb{R} \to \mathbb{R}$ be differentiable functions, such that

$$F'(x) = G'(x) \text{ for all } x \in \mathbb{R}.$$

Show that there exists a constant $C \in \mathbb{R}$ such that $F(x) = G(x) + C$ for all $x \in \mathbb{R}$. Note: Do this without integrating. In other words, work under that assumption that we do not yet know what integrals and anti-derivatives are.

1M: A valid approach to the proof is demonstrated, such as using the mean value theorem

1M: An appropriate structure for the proof is used (such as considering cases), that ensures the proof works for all x in R

1J: Conditions for using the Mean Value Theorem are checked

1J: Mean value theorem used correctly

1J: A correct and essentially complete proof

1N: Proof written clearly and easy to follow

Solution.
Let $h(x) = F(X) - G(X)$.
Since F and G are differentiable on $\mathbb{R}$, so is h.

<u>Case 1:</u> Let $x \in \mathbb{R}^+$.
Since h is differentiable on $\mathbb{R}$, h is differentiable on any $[0, x]$.
By the mean value theorem, there exists $c \in [0, x]$ such that

$$0 = h'(c) = \frac{F(x) - G(x) - F(0) + G(0)}{x}$$

So $F(x) = G(x) + C$, where $C = F(0) - G(0)$.

<u>Case 2:</u> Let $x \in (-\infty, 0)$.
It is similar, but we will repeat for completeness.
Since h is differentiable on $\mathbb{R}$, h is differentiable on any $[x, 0]$.
By the mean value theorem, there exists $c \in [x, 0]$ such that

$$0 = h'(c) = \frac{F(0) - G(0) - F(x) + G(x)}{-x}$$

So $F(x) = G(x) + C$, where again, $C = F(0) - G(0)$.

<u>Case 3</u>: Let $x = 0$.
When $x = 0$, $h(0) = F(0) - G(0)$.
The equation $F(x) = G(x) + C$ has $LHS = F(0)$, and $RHS = G(0) + F(0) - G(0) = F(0)$, so the equation $F(x) = G(x) + C$ holds for all $x = 0$.

Combining all cases, we have shown $F(x) = G(x) + C$ for all $x \in \mathbb{R}$, where $C = F(0) - G(0)$ is constant.